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GOVERNMENT OF INDIA
GEOLOGICAL SURVEY OF INDIA

INVESTIGATION FOR ORNAMENTAL GRANITE IN PARTS OF KOPPAL DISTRICT, KARNATAKA

(Code No: MIP/SR/KG/1999/007)

(Progress report for the field season 1999-2000)

By
Dr.S.PARANTHAMAN
And
T.BABU RAVINDRA KUMAR
Geologists (Sr)

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Project: Dimensional Stone
OPERATIONS: KARNATAKA & GOA

BANGALORE
2003

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STATUS OF MINERAL EXPLORATION REPORT IN TERMS OF UNFC COMPLIANCE	
Title of the report	Investigation for ornamental granite in parts of Koppal district, Karnataka FS: 1999-2000. Author/s: Dr. S. Paranthaman and T. Babu Ravindra Kumar. Acc. No. SRO-13363

Stage	**Nature of compliance required**	**Complied /Not complied (Page no)**	**Remarks**
1) Summary	(1) Factual Data	Complied in Abstract	
	(2) Recommendations	Not Complied	
	(3) Cost estimation	Not complied	
II) Introduction	(4) Scope & Purpose	Complied P.1	
	(5) Authority	Complied (FSP code)	
	(6) Current/Post Property Examination) BGI)	…..	
	(7) Principal Sources of Information	Complied in P.19	See references
	(8) Acknowledgement	Complied P.4	
III) Property Description	(9) Size, Continuity, Shape, surface rights	……..	
	(10) Third party issue	Complied (FSP code)	
	(11) Climate/rights with dates	Complied (FSP code)	
	(12) Land Title Officer	Complied (FSP code)	
	(13) Lat/Long/Village/town/ district/mine site	Complied P.20	See locality index
	(14) Accessibility/infrastructure	Complied P.2	
	(15) Land nature/water/timber/power/gas/agriculture/power	……	
	(16) Environmental issues/plan act/park/preservation	Not Complied	
	(17) Climate/topography	Complied P.2	

Stage	Nature of compliance required	Complied /Not complied (Page no)	Remarks
	(18) Impact of landscape on mining	……..	
	(19) Map of area/Block with roads/accessibility	Complied	
IV) History	(20) Previous work highlights	Complied P.2	
	(21) Attachment of previous maps/sections/tables	Not Complied	
	(22) Previous Mining History	Not Complied	
	(23) Reasons for closure	Not Complied	
	(24)Earlier production data	Not Complied	
	(25) Data on nearby mines	Not Complied	
(V) Geoscience Investigation			
(1) Geology	(26) Stratigraphy	Complied P.5	
	(27) Lithology	Complied P.5-8	
	(28)Structure (with maps/profiles)	……	
	(29) Alteration Zones	……	
	(30) Geotechnical	Complied P.15	
	(31) Adequacy of data density	……..	
	(32) Reliability (quality & quantity)	……..	
	(33) Geological model	Complied	
(II) Investigation methods	(34) Instrument & Technique	……..	
	(35) Geology (Map with scale)	Complied	Fig.1
	(36) Geophysical (with maps & scale)	……	
	(37) Geochemical (with map & scale)	…….	
	(38) Data density of above	……	
	(39) Size & style of mineralised zone	……	

Stage	Nature of compliance required	Complied /Not complied (Page no)	Remarks
	(40) Diagrams	……	
	(41) Metallurgical tests	……	
	(42) Deleterious contamination source	……	
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	(46) Reflectance of Mineralisation geology/gp/gc	……	
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(II) Sampling details	(48) Quality & Quantity of sample	Complied P.11	
	(49) Reliability on resource estimation	……	
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(V) Resource estimation	60) Geological interpretation	……	
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Stage	Nature of compliance required	Complied /Not complied (Page no)	Remarks
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	64) Estimation	Complied P.11	
(VI) Economic viability	65) Rough estimation of economic viability	……	
	66) intrinsic economic interest	……	
	67) Even tual economic exploration	……	

Report is in compliance with G-4 stage of mineral investigation. Different varieties of dimensional stones were demarcated. Resource was estimated. Hence report has been codified as 334 of UNFC.

Place: Hyderabad Date: Scrutinizer & Designation; Signature:

N. DEVARAJ
Director &RMH-11
Co-ordination
State Unit: Andhra Pradesh
Hyderabad-500068

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ABSTRACT

Investigation for resource evaluation of different types of ornamental stones associated with the Closepet granite and the adjoining Peninsular Gneissic Complex suite of rocks was taken up during the field season 1999-2000. As part of this programme, a total of about 1250 sq km was covered by geological mapping on 1:50,000 scale in parts of Koppal, Tavarakere and Gangavathi taluks of Koppal district, falling in Survey of India toposheets 57-A/2, A/5, A/6 and A/7. Besides identifying and demarcating the deposits of varieties of dimension stone granites, representative samples were collected from different granite types, for study of their petrographic and geochemical characters and also their engineering properties.

The area investigated in parts of Koppal district of Karnataka forms the northern extension of polyphase Closepet batholith. The rock types encountered in the area are the banded and foliated gneisses, migmatites belongs to Peninsular Gneissic Complex with small inclusions of mafic-ultramafic and metasedimentary rock assemblages equivalent to Sargur Group of rocks, volcano-sedimentary sequences of rocks form parts of the Kushtagi-Hungund schist belt of Dharwar Supergroup, a variant of post-Dharwar Closepet granite made up of pink and grey granite, either equigranular or porphyritic associated with pegmatite and quartz vein and a set of dolerite and gabbro dykes representing the younger post granitic intrusions.

The present work has brought out extension areas of premium and ordinary types of decorative granites exploited under the trade name multi-colour, Himalayan blue, NH Red, Chilly Red, Ikkal porphyry, etc, besides demarcating some of the new varieties such as Hasagal brown, Chocolate brown and Rangavathi grey.

The entire area of 1020 sq km of grey, pink and pink porphyritic granite and adjoining gneissic terrane were studied in detail as per the standard norms. A total recoverable reserves of the order of 39 million cubic metres was estimated from the Kanakagiri sector over a depth of 10 m from the ground level with a recovery of 20%.

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I. INTRODUCTION

Karnataka possess varieties of ornamental/decorative granites and has Century old tradition of using stones for various construction purposes. Its durability and availability have led to its wide and successful use as monuments and memorials.

The State is endowed with nearly one hundred types of decorative stones and most commonly known varieties are pink granite, multi-colour granite, black granite, grey granite, green granite and yellow granite. These rocks are estimated to contain more than 40% of the country's reserves of dimension stone and spread over an area of about 4262 sq km in the State.

Recognising the importance of these dimension stones in the national and international trade, systematic studies have been initiated in Geological Survey of India during 1994-95 to scientifically evaluate their quality and resource potential.

The present investigation has covered an area of 1250 sq km in parts of Koppal, Tavarakere and Gangavathi taluks of Koppal district, from where occurrences of premium equigranular and porphyritic red granite, grey and also the multi-colour varieties are identified. Some of the premium red granites are actively exploited in the area investigated. The area forms one of the important granite tract in northern Karnataka in Closepet suite of granitoids intruding into the PGC. The different commercial types available from these areas include the Ruby red, NH red, chocolate brown, Chilly red, etc. Multi-colour types are also identified associated with the PGC.

During this investigation, the assigned area was mapped on 1:50,000 scale and different commercial types vis-a-vis petrographic types were demarcated. The areas covered by commercial granite are further studied in terms of different defects such as sudden change of colour and texture of the rocks, shears, pegmatite veins, mafic segregation, migmatitic structures, etc.

i) Location and accessibility:

The area of investigation is encompassed between latitude $15^0$25'00"-$15^0$48'45"N and longitude $76^0$00'00"-$76^0$30'00"E and falls in the Survey of India toposheets 57-A/2, 5, 6 and 7 and forms parts of Yelberga, Koppal, Kushtagi and Gangavathi taluks of Koppal district. The area is connected by a network of motorable roads connecting important places like Raichur, Koppal and Bellary. The National High No. 13 connecting Bangalore-Sholapur passes in the eastern portion of the area. The other roads, such as Gangavathi-Koppal, Sindhanur-Kushtagi, Yelberga-Kushtagi, provides a good approach to the area under study. The nearest railhead is Koppal, situated on the Hubli-Guntakal metre-gauge section of the South Central Railway line and is about 40 km from the study area.

ii) Physiography and drainage:

The area examined is an undulatory terrain with a rugged topography covered by cluster of hillocks, knolls and tors occupied by the granite, gneisses and meta-volcanics. Besides, a chain of E-W to WNW-ESE trending hill ranges of Closepet granite running between Hire Banakal and Sulikere occupies the southwestern part of the area. In the central part between Somasugar and Katigihalli, WNW-ESE trending linear hills of migmatite, gneiss and granite and dykes are seen. These hill ranges in the southwestern part are seen under forest cover with fully dense scrub falling in the Agoli Reserved Forest range. The maximum height of the hill in this area is Δ712 m and is located south of Hamsadurga.

These hill ranges form the water sheds for the numerous nallas flowing west to east with dendritic drainage pattern.

The area falls in the semi-arid zones and experiences a very dry climate for most part of the year and receives very scanty rainfall with an annual rainfall of 65 cm. The water table is very shallow in the granitic country and vary in depth between 2 to 6 in. The water table goes down to a depth of 10 m during summer. About 40% of the area is covered generally by red soil and the remaining part is covered black soil. The red soil is invariably capping the pink granite and pink porphyritic granite, whereas the black soil is found covering the areas occupied by the grey granite and gneisses.

iii) Previous work:

Bruce Foote (1876) had referred to the granitic rocks of this area in his classical memoir entitled "The Geological Features of the Southern Maratta Country and Adjacent District". Subsequently officers of Hyderabad Geology Department

(1934-36) mapped a part of the area between Krishna and Tungabhadra Rivers. According to him, the grey and pink gneiss/granite with their associated acid members represents a portion of Peninsular Gneissic Complex.

Balasubramanian (1967-68) conducted first generation mapping in parts of 57-A/NW (parts of 57-A/5, 6 and 7) and described various types of rocks belongs to Dharwar and post-Dharwarian age. According to him, the metasedimentaries and the volcanics of the Kushtagi schist belt have undergone granitisation and have been intruded by profuse acidic and basic igneous rocks.

Venkoba Rao (1967-68) mapped the adjacent area and considers the granite and gneisses as part of the PGC and intrusive into the older metamorphites. He described the medium to coarse, porphyritic granite serve as beautiful decorative granite stone, when polished.

Ananthanarayana (1987-88), while conducting systematic geological mapping in the area, has given detailed account of the various lithounits and also furnished the geochemistry of the various granitoids in the area. He reported the availability of varieties of ornamental granite in the area.

iv) Present work:

In view of the growing demand for the Indian granites in the global market, so also in the domestic market as decorative/building materital, detailed study of granites has gained prominence in the field season programme of Geological Survey of India, study on resource evaluation of ornamental stone granites have been continuing in Operations Karnataka and Goa since field season 1994-95.

Granites and its variants including gneisses occupy more than 70% of the geographic extent of Karnataka State. The canvas of gneisses and migmatites, which occupy the entire State of Karnataka are intruded by a number of granitic bodies of different geo-chronological ages ranging from 2600 to 800 Ma. These lithological units posses valuable and pleasing coloured granite deposit, besides jet black dolerite dyke of premium quality. Keeping in view of this, commercial evaluation of these granites has been done in two phases spanning between 1994-97 and 1997-99. The first and second phases have brought to eight existence of premium commercial granite varieties in the State. Studies carried out during the second phase in parts of Koppal, Raichur and Bellary districts have revealed occurrences of several premium variety of grey and pink granite as well as gneisses. In view of this, occurrence of rich deposits of ornamental stone granite in the area, a two year programme is assigned with a main objective for systematically assessing the granites and gneisses and to demarcate the target areas for resource evaluation. The target areas are then studied in

detail to assess the deposits both qualitatively and quantitatively. The representative samples of the varieties identified are collected for determination of their petrological, physical and mechanical properties.

The present work includes detailed mapping on 1:50,000 scale to delineate the different commercial type of coloured granites and evaluation of their reserves. The area covered forms part of the Closepet suite of granitoids occurring in parts of Koppal district, which is the southern continuation of the world famous red granite tract in the vicinity of Ilkal area. The existing quarries were examined and studied in detail, the geological characteristics of these decorative stones, to finally evaluate the possible recoveries of the marketable blocks of stones from these deposits. The recoverable reserves of different commercial types were estimated for over a depth of 10 m from the ground level.

The quantum of work carried out is as follows:

Nature of work	**Total work envisaged for 1999-2000**	**Work proposed for each party**	**Work completed for FS 1999-2000 for two officers**
Large scale mapping on 1:50,000 scale (sq.km)	2500	1200	1250
SMPL (Nos)			
Petrological	25	13	20
Geochemical	50	25	10
Polishing	10	5	10
XRF	-	-	27

v) Acknowledgements:

The authors express their sincere thanks to Shri M.M.Swamy, Director, Project: Dimension Stone, Operations Karnataka and Goa for valuable suggestions in the field and scrutinising the report.

II. GEOLOGY

The area of investigation represents the northern extension of Closepet batholith. The rock types encountered in the area of operation are the banded and foliated gneiss, migmatite belongs to Peninsular Gneissic Complex (with small inclusions of mafic- ultramafic and metasedimentary assemblages equivalent to Sargur Group of rocks), volcanosedimentary sequences of the Kushtagi-Hungund schist belt of Dharwar Supergroup, a variant of post-Dharwar Closepet granite with associated pegmatite/quartz vein, and a set of dolerite and gabbro dykes representing the youngest part of granitic intrusion.

The stratigraphic position of different rock types in the regional geological framework along with their equivalent ornamental stone varieties are as follows:

Intrusives	Pegmatites and associated quartz veins Dolerite/gabbro
Closeper granite 2500-2700 Ma	Equigranular medium to coarse pink granite (ornamental-chilly red RBI red, Masal pink) Pink porphyritic granite (ornamental stone varieties; Ilkal porphyry etc) Medium to fine grained equigranular grey biotite granite (Ornamental stone varieties: Gangavathi grey, Mudgal grey)
Dharwar Supergroup	Meta-basalts and BMQ
Peninsular Gneissic Complex 2700-3000 Ma	Grey biotite gneiss Banded pink migmatite (Ornamental stone variety: Multicolour granite) Granodioritic gneiss (ornamental stone variety: Himalayan Blue type)
Sargur Group (?) >3000 Ma	Meta-pyroxenite, amphibolite and fuchsite quartzite and BFC

1. Sargur group:

The early supracrustal rocks equivalent to Sargur Group consists widely scattered rocks represented by meta-ultramafics. amphibolite, BFC and fuchsite quartzite occurring as unmappable bands within the granites and gneisses.

(a) Amphibolite occurs as rafts and xenoliths within the greywacke and gneisses in conformity with the general trend of the Closepet granite. The outcrop of this rock is seen mainly in the area east of Bhatnarasipur and Ghandhal. The rock is very fine grained, massive and foliated at places and traversed by very thin quartz veins.

(b) Meta-pyroxenite body cofolded together with gneisses occurs as remnants within the pink granite, with their fold axial traces along E-W direction. The outcrop of this rock is seen at 1 km north of Kannerumedu. Another meta-pyroxenite body is located in the area 0.5 km west of Noolihalli. The rock, in general, is very coarse grained, show dark grey to green in colour and is made up of coarse pyroxene and hornblende and with very little feldspar altered to sericite and permeated by criss-cross pink pegmatite.

(c) A NW-SE trending band of fuchsite quartzite intimately associated with mafic- ultramafic rocks is traced over 2 km discontinuously with an average width of 25 m. They form linear ridges within PGC located between 2 km NW of Kanakagiri and 1 km east of Wadagatti. The rock is highly schistose at the margin. The rock is fine grained, greenish in colour and is made up of quartz, fuchsite and muscovite.

2. Peninsular Gneissic Complex:

PGC is exposed in the central portion of the area, covered on either side by Closepet granite, occupying mainly the plain country and represented by the banded and foliated biotite gneiss, migmatite and granodiorite gneiss. Grey biotite gneiss is the well foliated rock unit, with mafic and felsic alternating layers and is made up of quartz, feldspar, biotite and hornblende. They are seen frequently permeated by the pink potash feldspar exhibiting multi-colour structure. They also occur as nebulitic bands and lenses within the Closepet granite. The pink banded gneiss and granodiorite gneiss at places shows attractive structural pattern akin to premium ornamental granites, being quarried elsewhere in the State under the trade name multi-colour and Himalayan blue granites. Prominent outcrops are seen near Sidhlabhavi, Hire Mukarthial and Hosegudda areas.

3. Dharwar Supergroup:

Meta-basalts and metasediments form part of Hungund-Kushtagi schist belt of Dharwar Supergroup. These rocks occur as linear belt in the northeastern corner of the area.

(a) Meta-basalt is the dominant rock unit in this group bordering Closepet granite and often occurring with tongues and apophyses of granite within them. The basaltic lava is greyish in colour, fine to medium grained, massive to schistose in nature. It composed of hornblende, plagioclase, actinolite, tremolite, chlorite and little quartz. Primary volcanic structures, such as vesicular and pillowed structures, are commonly seen in the area around Gudadur, Hosur, etc.

(b) Metasediments representing the banded magnetite quartzite are seen closely associated with meta-basalt.

4. Younger intrusives:

These granites that are generally included under Closepet granite are the most predominant type of rocks in the area of investigation. These include both grey and pink varieties with different textures. They hosts a variety of decorative granites. They form cluster of hills, knolls, tors and scattered bouldery outcrops aligned roughly in a NW-SE to WNW-ESE direction. This can be traced from Basapatna in the south to Gajendragarh intruding the volcano-sedimentary sequence of Dharwar and PGC in the north and southern portions respectively.

The Closepet granite in the area is a composite granite involving (multi-phase) two major phases, viz. pink and grey and a minor phase of mafic rich granite. The major phases exhibit both transitional and sharp contacts with each other. The grey granite, in general, formed by injection of granitic fluids into the Peninsular gneisses and subsequent K-metasomatism must have been caused the formation of pink porphyritic granite with the culmination of pink granitic phase. Mineralogically, these two major grey and pink phases are similar containing quartz-K-feldspar, plagioclase, biotite and hornblende.

The principal types of granites identified are fine to medium grained, equigranular grey granite, pink biotite granite, pink porphyritic biotite granite and medium to coarse grained equigranular pink granite.

(a) Grey granite is the major lithounit occupying the eastern portion of the area in a NW- SE passion covering an areal extent of 425 sq km (which is about 42% of the area occupied by the Closepet granitoid complex in the area). It occurs as isolated mounds, tors and knolls around Chickmadinal, Hiremadinal, Kanhal, Gudadur, Tavarakere, Bhalnarasapur, Venkatagiri, Basapatna and east of Kanakagiri. At many places, it is seen as caught up patches within the pink granite, indicating that this granite is older than pink granite. They grey granite is equigranular, fine to coarse grained, massive, hard and compact and consists of quartz, plagioclase and biotite. At many places, as seen in the area SE of Tavarakere, east of Gudadur and around Kalkere, these grey granites consist of lenses of pink porphyritic granite. These bodies are not found suitable for using as decorative granite because of uneven distribution of pink feldspar porphyry. The grey types, in general, are not suitable for commercial use. However, small lenses of fine grained grey granite occurs within the medium grained variants located 2 km east of Chickmadinal and 3 km east of Kanhal has the texture similar to the silver grey types, but bodies are very small and do not have wide areal extent to be qualified for commercial use. A small elliptical body of medium

grained equigranular grey granite with widely distributed feldspar grains, located 2 km SE of Venkatagiri, exhibit grey colour and uniform texture, which can be used as'ornamental granite for local industry.

(b) Pink porphyritic granites are extensively seen in the southwestern portion of the area under study. They occupy an area of nearly 590 sq km, which is 57.80% of the granite area. They occur as cluster of hillocks and bouldery outcrops. It is non-foliated granite, variably exhibiting porphyritic texture. The rock mainly composed of randomly oriented pink feldspar phenocrysts (2 to 5 cm long) set in a biotite rich matrix. Prominent outcrops are located in the Chicksulikere, Hosur, Adhivibhavi Tanda, Cheleri, Bommasagara Tanda. Vitalapura, Yoddoni, Nigalbandi, Basapur. Gunhal, etc. In the vicinity of basic dykes, the pink feldspar phenocrysts in the granite exhibit deep pink and red colours. At a few places, around eastof Adhivibhavi Tanda, SW of Chicksulikere, SW of Hosur, north of Bommasagara Tanda, the pink porphyritic granite consists very closely arranged phenocrysts of pink feldspar in a groundmass having both pink and grey feldspar and blue quartz with negligible amounts of biotite. This type of pink porphyritic variety occurs as small circular (or) elliptical bodies in the area and they are akin to the popular decorative granite used in the slate industry under the trace name Ilkal porphyry, Savan rose, Rosy pearl, Tumkur porphyry, etc., being mined elsewhere in the State. The commercial pink porphyritic granite reported near Chicksulikere and Hiresulikere areas has very largely widely distributed pink feldspar phenocrysts. The phenocrysts are dark brown, deep red or light pink in colours. The decorative grade of pink porphyritic granites are located near Hosur, Hagedal, Ningalbandi, etc. They consists of randomly oriented and closely arranged pink feldspar phenocrysts unlike Ruby red granite, thin variety consists of groundmass of little/more amount of white feldspar and biotite.

(c) There are two types of pink granites mapped in the area, via. (i) coarse grained pink granite and (ii) fine to medium grained pink biotite granite. These occur as isolated circular bodies within the vast stretch of pink porphyritic granite.

(i) The pink biotite granite occurs as small patches within the PPG, occupying an area of 1 sq km, which is about 0.10% of the total granite area. It occurs as small low lying hillocks and knolls near Hasagal, west of Ganganal, Akalkuppi and Uplapur. The pink biotite granite is massive, hard and compact and non-foliated. It is equigranular, medium grained, dominantly consists of equal amounts of pink and white feldspar, smoky or blue quartz and considerable amounts of biotite. At places, this granite consists of minor shears filled with green epidote. This granite may be tried commercially for local use, as it is uniform in colour and equigranular texture.

(ii) The medium to coarse grained granite occurs as lenses and bands within the pink porphyritic biotite rich granite. It occupies a total area of 4 sq km (0.4% of

the total granite) in the area of investigation. This granite occupies small isolated hills and knolls located near Venkatagudi. Arasinakere and Advibhavi. It consists of very closely knitted brick red colour potash feldspar with characteristic rounded to subrounded white feldspar grains. Their intergranular spaces are occupied by smoky or blue quartz and negligible amounts of mafic (biotite). In the area near Talakere, Katigehalli, Bommanahalli Tanda and Yettagal, this granite exhibits coarse grained texture without white feldspar grains. These types may represent the culmination product of major pink phase due to potash metasomatism in the Closepet granite. These varieties have little similarity to the premium commercial red granites quarried near Ilkal and Pattadkal areas and marketed under RBI red/NH red.

Pegmatite and quartz veins:

The granite and gneiss are traversed by many generations of pink and white pegmatite veins aligned in various directions. The brecciated one folded quartz vein forms WNW-ESE trending small ridges as seen south of Ghandal and NW of Gouripur.

Basic intrusives:

Several younger basic intrusives are seen in this area cutting the principal rock units. They trend generally in WNW-ESE and N-S to NNW-SSE directions. These dykes are generally gabbroic to doleritic in composition. They extend along the strike for several kilometers with a width varying from 25 to 60 m. These dykes occur at Hiremannapur, Kanhal, Wandagatti, SW of Gudadur, Advibhavi, Uchalgunti. etc. They are medium to fine grained, equigranular and show subophitic texture with augite and plagioclase feldspar as predominant minerals. Marginal uralitisation and sericitisation are commonly seen. These dykes are highly jointed and not suitable for quarrying into gangsaw size blocks for polishing industry but are quarried locally for rough building stones.

III. CRITERIA FOR SELECTION OF COMMERCIAL GRANITE

(a) Quality and quanty:

The red and grey ornamental granites demarcated in the investigated area are uniform in colour, grain size, texture and on polishing exhibits high reflecting. They are mostly devoid of undesirable physical characters, viz. shears, hairline cracks, basic inclusions, etc. These granites occur in huge volume in the investigated area.

(b) Marketing potential:

The red coloured granites, both porphyry and equigranular, located in the investigated area, besides serving as export oriented ornamental stones, these varieties have a good demand in local market/industry, especially for making wall panels and floor tiles.

(c) Engineering properties:

The different commercial varieties identified in the Kanakagiri sector possess the required engineering properties, such as (i) bulk specific gravity, (ii) water absorption, (iii) porosity, (iv) density, (v) compressive strength, (vi) tensile strength, (vii) youngs modules, (viii) primary and secondary wave velocity.

(d) Quarribility of the deposit:

The commercial granites-identified in parts of Koppal district have reasonably good reserves and are expected to produce gangue saw size blocks by selective quarrying. These deposits are surrounded by barren lands and can be developed with a network of road access upto the working site for the free movement of machinery.

(e) Rarity of occurrences:

The red coloured equigranular and porphyritic granite in the investigated area is equivalent to Ilkal red granites. Ilkal red variety is being quarried and marketed profitably as prime variety in both national and international markets.

IV. METHODOLOGY OF RESOURCE EVALUATION

The resource evaluation of different varieties of pink and grey granites and multicolour ornamental granites were made on the basis of systematic geological mapping and structural studies carried out in and around the quarries located in the area.

The following methodology have been evolved to study the grey and coloured granites in Kanakagiri sector:

(1) The premium and ordinary decorative granites introduced in the trade over two decades by the granite industry on the basis of its aesthetics, colour and other physical properties are taken as the standard and are delineated on the geological map along with the lithounits they are occurring.

(2) Potential areas are demarcated and further classified into different categories on the basis of their occurrence and physical parameters.

(3) The recoverable reserves for each decorative type were estimated on the basis of the existing ground realities and recovery factor obtained from working quarries at this area.

(4) The petrography, XRF analyses and engineering properties of the individual types were determined.

As seen in the area investigated, the strike length of individual deposits in 26 sub- areas ranges from 500 to 2500 m, with their width varying from 100 to 1250 m. All these deposits occur as low mounds ranging in heights between 5 and 20 m.

The area was mapped on 1:50,000 scale to delineate different types of prospective decorative granites. The area covered by decorative types is further studied in detail to evaluate marketable grade of material. The parameters usually evaluated area (a) uniformity of texture and colour, (b) jointing pattern that yield finally the gangue saw size (> 2 nT) and (c) weathering. The existing quarries were studied in detail to qualitatively evaluate the recovery of marketable grade material. Based on these properties, the areas have been divided into (i) defect free areas with good recoveries of marketable grade blocks and (ii) defective areas with poor recovery of blocks.

The total recoverable reserves of different commercial grey and pink- granite varieties from Kanakagiri sector are estimated to be 38.99 mm^3 at an average recovery of 20%. A depth of 10 m is considered below the ground level, while

calculating the reserve. The break-up of reserves under different commercial types are given in the following Table.

Table showing the reserves of different ornamental granites of Koppal district, Karnataka

Sl. No	Commercial Varieties	Lithounits	Areal extent (sq. Km)	Total reserves (mm^3)
1	Multi-colour	Pink migmatite (banded gneiss)	4.245	7.950
2	Himalayan Blue	Granodiorite gneiss with veins of pink pegmatite	2.000	4.750
3	NH Red Ilkal types	Coarse grained pink granite	1.000	2.180
4	Chilly red/ RBI red	Coarse grained equigranular granite	2.100	7.440
5	Chocolate brown (Hasagal brown)	Medium grained biotite rich pink granite	0.900	2.945
6	Ilkal porphyry (Hosur and Sulikere Radiant Red porphyry)	Pink porphyritic granite	4.500	9.750
7	Mudugal grey (Gangavathi grey)	Medium to coarse biotite rich grey granite	0.500	2.100
8	Tavarakere silver grey	Fine grained equigranular granite	0.600	1.875
TOTAL			**15.840**	**38.990**

V. DESCRIPTION OF PROSPECTIVE AREA

1. Kanakagiri sector:

The present investigation covers the northern part of the Closepet granite around Kanakagiri, Arasinakere, Hosur, Hiresulikere, Huliyapur, Huliraider, Devalaprr and Hadegal areas in parts of Yelbarga, Kushtagi, Koppal and Gangavathi taluks of Koppal district, Karnataka, where isolated bodies of both pink, grey and multi-colour varieties of commercial granites are identified.

During the course of survey, the granite and gneisses in Kanakagiri sector, covering a total of 1250 sq km, have been thoroughly scanned and different types of prospective decorative granites are identified.' The Kanakagiri sector has been further divided into several sub-sectors. The commercial varieties of Dimension stones are categorised on the basis uniformity of colour/texture, joint pattern, defects, alterations,

thickness of weathering/overburden and the ability to yield finally the gangue saw size blocks.

The Kanakagiri sector is the southern extension of the world famous Ilkal ornamental red granite tract, it covers an area of 1250 sq km and enclosed between latitudes $15^0$25'00" - $15^0$48'45" and longitudes $76^0$00'00"M - $76^0$30'00".

The variants of the Peninsular Gneissic Complex and Closepet granites are the main source types generating varieties of ornamental/decorative granites in this sector.

Varieties of ornamental granites like multi-colour and Himalayan blue are associated with PGC, whereas variants of pink and grey granites showing both porphyritic and equigranular textures are associated with Closepet granites. These popular varieties occur as small isolated bodies in this sector and are quarried at places for both export and domestic use. These granites are commercially termed equivalent to some of the well-established ornamental granites being quarried and marketed elsewhere in the State under trade names multi-colour, Himalayan blue, Ilkal porphyry, NH red, Chilli red, Chocolate brown etc.

Description of individual types in Kanakagiri sector

(A) Dimensional stone associated with PGC:

(i) Himalayan blue:

This is a commercial name given to a pink migmatite of dioritic composition. It shows dominant bluish grey colour of. veins of pink feldspar injection. The granites of this variety is being quarried extensively around Kukkanur area, Koppal district. It is an additional deposit spread over 2 sq km with a reserve of 4.75 mm^3. Workable deposits are identified south of Hire Mukarthial and SW of Siddapur areas. The average sized blocks quarried in this area vary from 2 to 3 m^3.

Engineering properties of a sample collected from the area near Hire Mukartihal has given specific gravity 2.65; water absorption ratio 0.04%, porosity 0.16%, density 2.57 gm/cc and compressive strength 596.7268 kg/cm2. The above properties indicate that the rock is very hard and tough and suitable for using as ornamental/decorative stones. The water absorption ratio and porosity are also very low. The chemical properties of this variety is described under Chapter "Geochemistry".

(ii) Multi-colour:

Multi-colour in general is the commercial name given to a group of pink migmatites, made up of pink coloured feldspathic injection into the grey biotite gneiss, showing different degrees of assimilation produced by the Closepet granite activity. These migmatites with pleasing shades show uniform pink colour in combination with a variety of migmatite structures. The rock is light to medium pinkish in coloud with fascinating pink and smoky grey bands. The rock is foliated, medium to coarse grained and consists of predominantly quartz, potash feldspar and biotite. Biotite occurs as streaks, bands, schlieren and also as nebule. Microsection studies of both Himalayan blue and multi-colour commercial granites generally show equigranular, subhedral and hypidiomorphic texture with gneissic banding and chiefly composed of quartz, orthoclase, plagioclase, biotite and hornblende. The epidote and opaque iron ores formed the accessory minerals and chlorite and sericite constitutes the secondary minerals. The quartz is euhedral shows undulose extinction indicating the effects of deformation. The plagioclase is subhedral colourless and exhibits polysynthetic twinning, show alteration effects to sericite indicative of retrograde effects of metamorphism. A few biotite grains altered to chlorites indicate that the rock is easily processable in terms of polishability. The distribution of this types are identified (i) 750 m south of Gorlahalli, (ii) NW of Magnekal, (iii) 1 km SE of Siddapur, (iv) 1.5 km SSW of Gandhal, (v) west of Negalpura and (vi) around Horegud. These deposits are not of export quality, but can be selectively utilised for the local use. The total area covered by this deposit is around 4.245 sq km. A total reserve of 7.95 million cubic metres was estimated upto a depth of 10 m from the ground level considering 20% recovery factor.

Engineering properties in respect of two samples collected from the above areas are tabulated below:

Location	**SE of Hosagudda**	**SW of Siddapur**	**Average**
Sample	(1)	(2)	
Specific gravity	2.64	2.62	2.63
Water absorption	0.10	0.07	0.085
Porosity	0.26	0.18	0.22
Density gm/cc	2.59	2.57	2.58
Compressive strength kg/cm^2	1680.183	1240.531	1460.657
Tensile strength mg/cm^2	68	82	75
Vs km/sec	5.00	5.70	5.35

The properties indicate had and tough nature with low porosity and water absorption ratio.

(B) Dimenstion stone associated with Closepet granite:

(i) Chocolate brown (Hasagal brown):

This commercial variety is the medium grained, equigranular biotite rich granite, consisting of both white and pink feldspars and smoky quartz. The feldspars show chocolate brown to purplish colour and occur as isolated pockets within the pink porphyritic granite. This is the new variety of ornamental stone granite identified during the present work. The rock is massive and less weathering. It is also devoid of any significant defects, except for the presence of minute fracture filled with green epidote at places. Workable deposits of this varieties are identified, 1) just north of Hosagal, 2) 1 km west of Ganganhal and 3) around Uplapur areas. The total area covered by this type is 0.90 sq. km and estimated to contain a reserve of 2.945 million cu. m.

Engineering properties of a sample collected from the area near Hosagal has given specific gravity 2.62, water absorption ratio: 0.12%, porosity 0.31%, compressive strength 1888.569 Kg/cm^2 and tensile strength 68 Kg/cm^2. The above properties indicate that the rock is very hard and tough and suitable for using for ornamental/decorative purposes. The water absorption ratio and porosity are also very low. The chemical properties/components of this rock are described under chapter "Geochemistry".

(ii) Chilly Red (Arasinakere Red):

This is the commercial granite given to coarse grained granite showing chilly red/brick red colour. It shows equigranular texture and consists of small grains of rounded white and large grains of brick red feldspar, white quartz and small components of mafic. This commercial granite is the premium variety, equivalent to "Chilly red" quarried near Pattadkal and Mangalagudda areas in Bagalkot district with the marked difference of the presence of characteristics rounded milky white feldspar grains in the former.

Reports of this varieties occur as isolated bodies within the pink porphyritic granite and are demarcated near Arasinakere and south of Adhivibhavi. They are spread over an area of about 2.10 sq km with an estimated probable reserves of about 7.44 million cubic meters.

Engineering properties of a representative sample collected west of Arasinakere village has given specific gravity of 2.60, water absorption ratio 0.27%,

porosity 0.21%, compression strength 2927.282 kg/cm2, tensile strength 61.00 kg/cm^2. These indicated that the rock is hard and tough and can be used as building/decorative stones.

(iii) Ilkal pink porphyry/Radient red (Sulikere pink porphyry, Hosur red porphyry):

The pink porphyry types examined in the area are generally very coarse grained rock, made up of large phenocrysts of potash feldspar ranging in colour from pink, through red to dark reddish brown embedded in the groundmass of pink and white coloured feldspars, quartz and biotite. Petrographically, this variety is very closely resembles the porphyritic varieties like Savan rose, Tumkur porphyry and Ilkal porphyry. These units occur as lensoidal and circular bodies within the pink porphyritic granite. About seven such bodies are seen distributed in parts of Toposheets 57-A/2, 6 and 7. covering an areal extent of 4.50 sq km. These deposits occur in the following localities:

1. Southwest of Chik Sulikere
2. East of Hire Sulikere
3. Southwest of Hosur
4. North of Basapur
5. South of Hagidal
6. Southwest of Biraldine
7. West of Gunhal.

The deposits identified/demarcated near Hire Sulikere, Chick Sulikere and Hagidal areas exhibit dark brown colour and have huge feldspar phenocrysts, embedded with the groundmass of grey quartz and biotite. These rocks show crude alignment roughly along the general trend of the granite formation. The porphyritic types located SW of Hosur, exhibit blood red and tomato red colours and the phenocrysts are randomly oriented with negligible amount of biotite.

These varieties show comparatively less defects and are quarried occasionally. A total reserve of 9.75 million cu m was estimated from the above locations.

Engineering properties in respect of four samples collected from the above areas are tabulated below:

Locations	**West of Hire Sulikere**	**SW of Hosur**	**SSW of Hosur**	**Hagidal**
Properties	**1**	**2**	**3**	**4**
Specific gravity	2.63	2.61	2.60	2.64
Water absorption %	0.18	0.13	0.35	0.06
Porosity %	0.47	0.33	0.45	0.15
Density	2.62	2.78	2.61	2.60
Compressive strength kg/cm^2	1286.509	1191.882	598.4425	749.2729
Tensile strength kg/cm^2	107	91	75	75

These properties indicate that this rock is hard, tough with less porosity and water absorption ratio. They are well suited for using as decorative stone/building material.

(iv) Gangavathi grey:

This grey granite is medium grained, equigranular rock, consisting feldspar, quartz and wide distribution of biotite. The potash feldspar exhibits very light colour in the polished section, which is added attraction to this rock to be used as decorative stone. It looks slightly similar to the Mudgal grey commercial granite, being quarried extensively in the northern extension of the present area. This deposit form an isolated body, occurring as lensoidal unit within the grey granite, which is reported south of Venkatagiri in parts of Toposheet 57-A/7. The total area covered by this deposit is 0.50 sq km and an approximate reserve of 2.10 million cu m is estimated.

One sample is analysed for geotechnical properties. The results are as follows:

Specific gravity 2.63, water absorption ratio 0.10%, porosity 0.26%, compressive strength 1002.767 gm/cm^2, tensile strength 99 gm/cm^2.

These properties indicate that this rock is suitable for using as decorative stones.

(v) Tavarakere grey (similar to Silver grey):

This is fine grained, silver grey in colour with uniform colour and texture. It is similar to silver grey commercial variety. This deposit occurs as small isolated bodies within the biotite rich grey granite. About 7 such bodies are identified to the north and NW of Kanhal and west of Chick Mudinal areas. Quarrying of this granite can be

resorted only for local use and high sized gangue sized blocks cannot be quarried due to presence of close-spaced joints. A total reserve of 1.875 million cu m is estimated upto a depth of 10 m from the surface with an average recovery of 20%.

Petrography of Dimension stone associated with Closepet granites:

(i) Pink equigranular and porphyritic commercial types:

The thin sections of pink equigranular and porphyritic commercial granites are generally characterised by medium to coarse grained inequigranuiar and hypidiomorphic in textures. The sections, essentially consists of quartz, orthoclase, plagioclase, microcline, biotite and accessories being sphene, apatite and a few zircons. The orthoclase feldspar is subhedral forms phenocrysts as well as groundmass. It shows effects of sericitisation in a few sections. Microcline crystals are occasionally seen with iron hatched twinning. The albite is subhedral and shows carlsbad twinning. The quartz grains are mostly unhedral in shape and rarely subhedral with irregular grain boundaries. A few quartz grains show undulose extinction. Biotite, the major mafic constituent, occurs in subhedral form and exhibit very strong pleochroism from dark brown to green colour, and rarely show effects of alteration into chlorite indicative of retrograde metamorphism.

(ii) Grey commercial granite:

The grey commercial granite, under microscope, shows hypidiomorphic texture, essentially made up of subhedral plagioclase and hornblende and accessories are apatite and iron ore. Feldspars are sericitised in some cases, inclusions of quartz are seen in both orthoclase and plagioclase feldspars. Biotite constituting the major mafic minerals show inequigranuiar, strong pleochroic and rare alteration to chlorite.

The petrographic studies indicate that the pink commercial granites are fresh, unaltered, uniform and has close mineral mosaics. These rocks can also take good polishing.

Geochemistry:

For classifying various types of granitoids and also to obtain an insight into their chemical variations, the granites and gneisses of Koppal district, being used as ornamental purposes, have been sampled and analysed for major elemental composition using XRF at the PPOD Laboratory, GSI, AMSE Wing, Bangalore.

Table-1 presents the major elemental data in respect of various types of granites of Closepet batholith and Peninsular Gneissic Complex of Koppal district.

The pink equigranular, porphyritic and grey granite of Closepet granites range in SiO_2 composition from 71.27 to 74.23% (av. 72.66) and AL_2O_3 from 14.03 to 15.42% and the pink migmatitic grains of PGC range in SiO_2 composition from 71.55 to 73.32%. In both granites and gneisses, the ferromagnesian contents are typically low (FeO + MgO + TiO_2) with very low MgO (av. 0.33%).

In the normative An-Ab-Or (Fig. la) triangular diagram (O'Connor, 1965), most of the gneisses of PGC plot in the trondhjemite field, whereas the granites plot in the granitic field.

In the K-Na-Ca diagram (Fig. lb), the Closepet granites show quartz-monzonite composition, whereas the gneisses clustering on the tie line between tonalite-granodiorite field.

In AFM triangle, the granites and gneisses show a calc-alkaline trend drawn by Kuno (1968). Pink migmatite/gneiss clustering on the most fractionated extremity of the trondhjemite trend (Arth et al, 1978).

Major elements corroborate distinct character of the granites and gneisses. The gneisses represent an earlier phase compared to granite and display all features typical of Archaean TTG (Tonalite-Trondhjemite), whereas the granites of the Closepet batholith show granodiorite and granite phases and both generally alkali rich display more typical calc-alkaline affinities.

The normative composition indicates the presence of high alkali and silica percentages in both granites and gneisses.

VI. SUMMARY AND CONCLUSIONS

A total area of 1250 sq km was mapped on 1:50,000 scale, covering parts of Koppal district, falling in Toposheets 57-A/2, 5, 6 and 7. In addition, samples were collected from different variants of granite representing the varieties of ornamental stones for petrographic and XRF studies and also to evaluate their engineering properties and polishing characteristics.

The area investigated represents the northern extension of Closepet granite occurring in association with other rock types like amphibolites, meta-pyroxenites and quartzites belongs to ancient supracrustal (Sargur Group), band and foliation grey gneiss, migmatites and granodiorite gneiss of PGC. Volcano-sedimentary sequences of rocks represented by the Dharwar Supergroup and a set of dolerite dykes representing the youngest post-granitic intrusion. Closepet granite and the PGC are

the most predominant rock types in the area, hosting a variety of premium and ordinary types of decorative granites. Thus all the varieties identified in this area are the product of pink migmatite and grey or pink granite (both equigranular and porphyritic).

The different units of Closepet granite and the associated decorative granites have been delineated and assessed of their potentials. The area has a vast stretch of pink granite, pink porphyritic granite and pink migmatite, with varieties of pleasing colours, shades and textures yielding varieties of ornamental granites, like multi-colour, Himalayan blue, NH red, Chilly red and Ilkal porphyry, etc., which have established the brand names in both export and local markets.

During the course of survey, a total of 16 sq km area comprising varieties of commercial granite/ornamental stone granite have been mapped in Kanakagiri sector. The total research reserves of all the dimension stone granite varieties from these sector is estimated to be of the order of 39 million cubic meters at an average recovery of -20%. The mineralogical, petrographic, geochemical and geotechnical properties of these varieties indicate that these rocks are suitable to use as decorative/ornamental stone granites and can take good polishing.

Approved by
Sd/-
(M.M. Swamy)
Director,
Project: Dimension Stone Operations
Karnataka and Goa,
Geological Survey of India,
Bangalore.

Signature of authors
Sd/-
(S. Paranthaman)
Geologist (Sr)
Sd/-
(T. Babu Ravindra Kumar)
Geologist (Sr)

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LOCALITY INDEX

Locality	Latitude				Longitude			T.S. No.
Akalkuppi	15	37	20		76	29	32	57-A/06
Arasanakere	15	30	05		76	17	40	57-A/06
Adhivibhavi	15	35	15		76	20	45	57-A/06
Adhivibhavi Tanda	15	34	50		76	20	00	57-A/06
Bommasagara Tanda	15	30	55		76	18	50	57-A/07
Basapurf	15	41	00		76	07	30	57-A/02
Biraldinni	15	40	20		76	11	10	57-A/02
Bhat Narasipur	15	32	40		76	29	45	57-A/06
Basapatna	15	28	40		76	29	20	57-A/07
Chellari	15	34	40		76	16	20	57-A/06
Chick Madinal	15	31	15		76	22	45	57-A/06
Chick Sulikere	15	31	38		76	16	45	57-A/06
Ganganhal	15	32	45		76	18	30	57-A/06
Gauripur	15	36	50		76	21	10	57-A/06
Gorlahalli	15	42	50		76	19	05	57-A/06
Gunhal	15	35	30		76	13	40	57-A/02
Gandhal	15	39	15		76	19	08	57-A/06
Gudanur	15	38	45		76	28	00	57-A/06
Hire Sulikere	15	31	38		76	15	45	57-A/06
Hagedal	15	41	10		76	03	20	57-A/02
Hasagal	15	31	40		76	17	45	57-A/06
Hire Mannapur	15	45	50		76	16	45	57-A/05
Hire Banekal	15	26	20		76	25	27	57-A/07
Hamsadurga	15	28	50		76	21	05	57-A/07
Hire Mukarthial	15	42	55		76	22	00	57-A/06
Hosagudda	15	39	30		76	24	50	57-A/06
Hosur	15	31	15		76	16	10	57-A/06
Hire Madinal	15	30	38		76	23	22	57-A/06
Kanhal	15	42	25		76	26	40	57-A/06
Katigihalli	15	37	20		76	17	15	57-A/06
Kannerumedu	15	33	00		76	21	00	57-A/06
Kanakagiri	15	34	00		76	25	12	57-A/06
Kalkere	15	33	45		76	29	15	57-A/06
Mettagal	15	29	40		76	16	05	57-A/07
Magnekal	15	41	50		76	20	55	57-A/06
Nigalbandi	15	40	50		76	08	30	57-A/02
Nagalpura	15	33	00		76	22	10	57-A/06
Noolihalli	15	46	40		76	19	50	57-A/05

Locality	**Latitude**				**Longitude**			**T.S. No.**
Ramdurg	15	30	35		76	20	20	57-A/06
Siddapur	15	42	40		76	22	10	57-A/06
Somasagar	15	34	30		76	21	40	57-A/06
Sidhlabhavi	15	42	22		76	18	12	57-A/06
Talkere	15	39	45		76	15	35	57-A/06
Tavarakere	15	45	00		76	24	40	57-A/05
Uplapur	15	33	10		76	17	30	57-A/06
Uchalunti	15	37	00		76	15	10	57-A/06
Vitalapura	15	29	25		76	24	35	57-A/07
Venkobagudi	15	16	05		76	17	00	57-A/06
Venkatagiri	15	27	47		76	26	50	57-A/07
Wadagatti	15	38	30		76	20	50	57-A/06
Yoddaoni	15	41	20		76	14	15	57-A/02
Yelbarga	15	37	00		76	00	45	57-A/02

Table-1 : Major elements concentrations of granites (Closepet) and gneisses (PGC) of Koppal area, Karnataka (Oxides in Wt%)

Sample No.	PB-01	PB-02	PB-03	PB-04	PB-05	PB-06	PB-07	PB-08	PB-09	PB-10	PB-11	PB-12	PB-13	PB-14	PB-15	PB-16
SiO_2	72.26	71.47	72.80	71.99	72 88	71.27	73.32	72.55	72 58	72.21	74.23	72.98	70 30	71.55	71.51	73.30
Al_2O_3	13.42	13.89	13.79	13.97	13 64	13.95	14.59	15.22	15 42	14 08	13.22	13.68	14.92	14.03	13.83	13.28
Fe_2O_3	1.72	2.20	1 27	1.60	1 19	2.07	1.47	1.04	0.66	1.67	1.24	1.41	2.80	2.35	2.12	1.88
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MgO	0.26	0.39	0.20	0.21	0 27	0.27	0.32	0.25	0.07	0 22	0.05	0.23	0.84	0.67	0.40	0.22
CaO	1.21	1.52	1.30	0.81	1.27	1.32	1.85	1.46	1.76	1.32	0.85	1.32	3.02	2.12	1.39	1.61
Na_2O	3.66	3.95	3.73	3.31	3.59	3.90	5.68	5.48	5.36	3.42	3.64	4.06	5.23	4.75	3.92	4.75
K_2O	5.29	4.59	4.98	5.93	5.27	5.02	1.56	2.92	2.74	5.30	4.78	4.28	1.35	2.60	5.07	3.28
TiO_2	0.23	0.30	0.14	0.39	0.17	0.26	0.19	0.15	0.08	0.19	0.08	0.15	0.36	0.25	0.29	0.17
P_2O_5	0.05	0.09	0.05	0.06	0.05	0.07	0.05	0.04	0.03	0 07	0.02	0.06	0.11	0.07	0.07	0.05
MnO	0.03	0.03	0.01	0.02	0.01	0.04	0.02	0.02	0.01	0 02	0.02	0.03	0.03	0.03	0.03	0.03
CO_2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0 00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fi	0.02	0.02	0.03	0.02	0.02	0.01	0.03	0.02	0.02	0 04	0.01	0.03	0.00	0.01	0.01.	0.01
S	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0 00	0 00	0.00	0.00	0.00	0.00	0.00	0.00
Cr_2O_3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0 00	0 00	0.00	0.00	0.00	0.00	0.00	0.00
NiO	0.07	0.07	0.06	0.06	0.06	0.06	0.02	0.05	0.14	0.08	0.07	0.08	0.04	0.05	0.07	0.12
SrO	1.07	0.82	0.94	1.19	0.87	0.82	0.56	0.32	0.62	0.75	0.38	0.70	0.51	0.65	0.69	0.59
LOI	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total	**99.23**	**99.34**	**99.30**	**99.56**	**99.29**	**99.06**	**99.66**	**99.52**	**99.49**	**99.37**	**98.59**	**99.01**	**99.51**	**99.13**	**99.40**	**99.29**

Sample No.	PB-01	PB-02	PB-03	PB-04	PB-05	PB-06	PB-07	PB-08	PB-09	PB-10	PB-11	PB-12	PB-13	PB-14	PB-15	PB-16
K_2O/Na_2O	1.45	1.16	1.34	1.79	1.47	1.29	0.27	0.53	0.51	1.55	1.31	1.05	0.26	0.55	1.29	0.69
FeO(T)/FeO	0.88	0.86	0.88	0.89	0.83	0.89	0.84	0.82	0.91	0.89	0.96	0.87	0.79	0.80	0.85	0.90
CaO/Na_2O	0.14	0.18	0.15	0.09	0.14	0.15	0.26	0.17	0.22	0.15	0.10	0.16	0.46	0.29	0.15	0.20
A/NK	1.14	1.21	1.20	1.18	1.17	1.18	1.32	1.25	1.31	1.24	1.18	1.21	1.48	1.32	1.16	1.17
A/CNK	0.96	0.98	0.99	1.05	0.98	0.98	1.01	1.03	1.03	1.02	1.04	1.00	0.96	0.97	0.96	0.93
Quartz	27.60	27.18	28.62	26.88	28.68	25.80	29.34	25.74	26.52	28.14	32.88	29.64	27.36	28.56	25.74	29.82
Orthoclase	31.69	27.24	£9.47	35.03	31.14	30.02	9.45	17.24	16.12	31.69	28.36	25.58	7.78	15.57	30.02	19.46
Albite	30.92	33.54	31.96	28.30	30.39	33.54	48.21	46.64	45.59	29.34	31.44	34.58	44.54	40.35	33.54	40.35
Anorthite	4.73	6.67	6.12	5.84	5.84	5.56	9.73	8.06	10.01	7.23	5.28	6.39	13.34	9.45	5.00	5.28
Wollastoni	0.70	0.00	0.46	0.00	0.23	0.46	0.00	0.00	0.00	0.00	0.00	0.12	0.00	0.00	0.12	1.28
Diopside	1.51	1.08	1.08	0.43	1.51	1.51	0.22	0.00	0.43	0.86	0.00	1.30	0.86	1.73	2.16	1.08
Hypersthene	0.00	0.50	0.00	0.30	0.00	0.00	0.70	0.60	0.00	0.10	0.10	0.00	1.70	0.90	0.00	0.00
Olivine	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nepheline	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Leucite	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Aamite	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0 00	0.00	0.00	0.00	0.00	0.00	0.00
Rutile	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
llmenite	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.30
Chromite	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Magnetite	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Haematite	1.76	2.24	1.28	1.60	1.28	2.08	1.44	1.12	0.48	1.76	1.28	1.44	2.88	2.40	2.08	1.92
Apatite	0.00	0.34	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Corundum	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Sample No.	**PB-01**	**PB-02**	**PB-03**	**PB-04**	**PB-05**	**PB-06**	**PB-07**	**PB-08**	**PB-09**	**PB-10**	**PB-11**	**PB-12**	**PB-13**	**PB-14**	**PB-15**	**PB-16**
Other-Sill	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Other-Oxid	0.47	0.59	0.28	0.86	0.28	0.20	0.47	0.08	0.08	0.28	0.00	0.28	0.78	0.39	0.59	0.00
Diff. Ind.	90.21	87.96	90.05	90.21	90.21	89.36	87.00	89.62	88.23	89.17	92.68	89.80	79.68	84.48	89.30	89.63
Sof.lnd.	4.43	5.70	3.75	3.81	5.17	4.15	4.18	3.61	1.13	4.00	1.00	3.93	9.10	8.35	5.94	3.13
Total of Nor	**99.37**	**99.38**	**99.27**	**99.24**	**99.34**	**99.17**	**99.56**	**99.65**	**99.61**	**99.40**	**99.59**	**99.32**	**99.59**	**99.35**	**99.25**	**99.49**

1, 2, 3, 4, 5,10, 11, 15,16 - Pink equigranular and porphyritic granite (Closepet), 6, 12, 13 - Grey granite (Closepet)

7, 8, 9, 14 - Pink migmatitic gneiss (PGC), Major oxide by PW 1410 Phillips Plasmaquad by GSI, AMSE Lab, Bangalore.

Details of ornamental granites and their estimated reserves from different sub-areas in Kanakagiri sector in parts of Koppal district, Karnataka for the Field Season 1999-2000

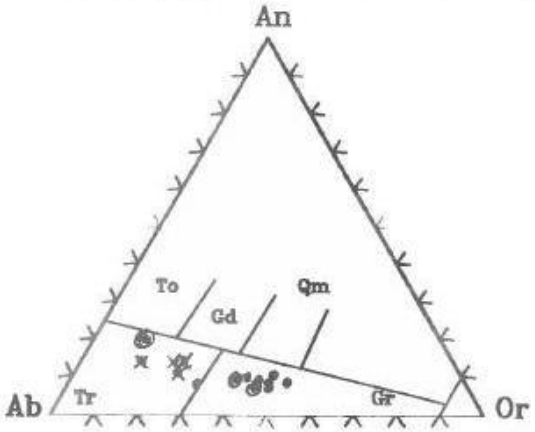
Sl. No.	Localities	Toposheet	Rock type	Commercial type	Strike (m)	Width (m)	Thickness (m)	Reserves below 10 m from surface (mm^3)	Reserves above the surface (mm^3)	Total Reserves (mm^3)
1	Gorlehalli	57 A/6	Pink migmatite	Multi colour	2000	500	10	2.000		2.000
2	Megnakal	57 A/6	Pink migmatite	Multi colour	750	250	12	0.355	0.095	0.450
3	Hosagudda	57 A/6	Pink banded gneiss	Multi colour	1500	1250	12	3.750	0.750	4.500
4	Akalkuppi	57 A/6	Pink migmatite	Multi colour	500	100	10	0.100		0.100
5	Gandhal	57 A/6	Pink migmatite	Muiti colour	1500	250	10	0.750		0.750
6	Nagalpura	57 A/6	Pink migmatite	Multi colour	750	100	10	0.150		0.150
7	Hire Mukarthial	57 A/6	Pink granodiorite gneiss	Himalyan blue	2500	750	12	3.750	0.750	4.500
8	Ramdurg	57 A/7	Pink granodiorite gneiss	Himalyan blue	500	250	10	0.250		0.250
9	Talakere	57 A/6	Coarse pink granite	NH Red	1500	150	10	0.450 i		0.450
10	Katigahalli	57 A/6	Coarse pink granite	NH Red	2000	175	15	0.700	0.350	1.050
11	Bommanahalli Tanda	57 A/7	Coarse pink granite	;NH Red	700	200	10	0.2801		0.280
12	Mettagal	57 A/7	Coarse pink granite	NH Red	1000	200	10	0.400		0.400
13	Advibhavi	57 A/6	Pink medium granite with white feldspar	Chilly/RBI Red	1200	500	12	1.200	0.240	1.440

Sl. No.	Localities	Toposheet	Rock type	Commercial type	Strike (m)	Width (m)	Thickness (m)	Reserves below 10 m from surface (mm^3)	Reserves above the surface (mm^3)	Total Reserves (mm^3)
14	Arasinakere	57 A/6 & 7	Pink medium granite	Chilly red	2000	750	20	3.000	3.000	6.000
15	Akalkuppi	57 A/6	Pink biotite rich granite	Chocolate brown	1500	200	12	0.600	0.120	0.720
16	Uplapur	57 A/6	Pink biotite granite	Chocolate brown	750	150	10	0.225		0.225
17	Hasagal	57 A/6	Pink biotite granite	Chocolate brown	1000	500	20	1.000	1.000	2.000
18	Chick Sulikere	57 A/6	Pink porphyritic granite	Ilkal porphyry type Sulikere porphyry	2000	350	15	1.400	0.700	2.100
19	Hire Sulikere	57 A/6	Pink porphyritic granite	Sulikere porphyry	1000	250	12	0.500	0.100	0.600
20	Hosur	57 A/6	Pink porphyritic granite	Hosur porphyry	2000	350	15	1.400	0.700	2.100
21	Basapur	57 A/2	Pink porphyritic granite	Ilkal porphyry	1000	300	12	0.600	0.120	0.720
22	Hagidal	57 A/2	Pink porphyritic granite	Ilkal porphyry	1000	750	12	1.500	0.300	1.800
23	Biraldine	57 A/2	Pink porphyritic granite	Ilkal porphyry	1000	700	12	1.400	0.280	1.680
24	Gunhal	57 A/2	Pink porphyritic granite	Ilkal porphyry	750	500	10	0.750		0.750
25	Kanhal	57 A/6	Fine grey equigranular granite	Silver grey	2500	250	15	1.250	0.625	1.875

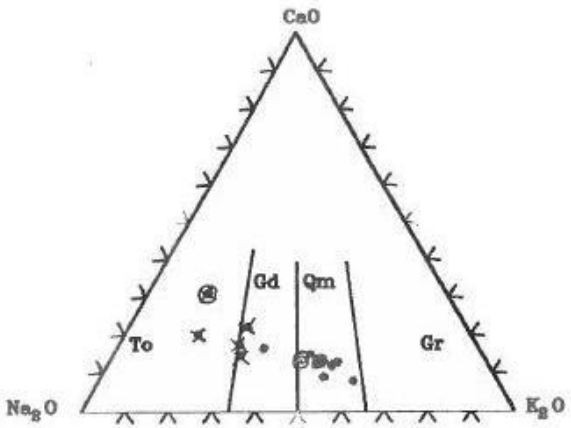
Sl. No.	Localities	Toposheet	Rock type	Commercial type	Strike (m)	Width (m)	Thickness (m)	Reserves below 10 m from surface (mm^3)	Reserves above the surface (mm^3)	Total Reserves (mm^3)
26	Venkatagiri	57 A/7	Coarse grey biotite granite	Mudgal type	1500	350	20	1.050	1.050	2.100
	Total							28.810	10.180	38.990

*1. Thickness assumed 10 m from below the surface
*2. Reserve computed with 20% recovery

Figer

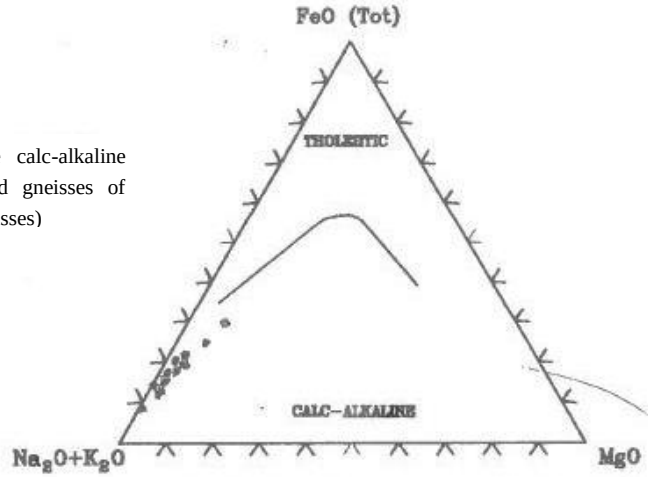


Normative An-Ab-Or Diagram for granite and gneisses of Koppal area (O Connor 1965)



K-Na-Ca Triangle for granite and gneisses from Koppal area. Barker and Arth (1976)

AFM Diagram showing the calc-alkaline Character of the granite and gneisses of Koppal area (Granites and gneisses)



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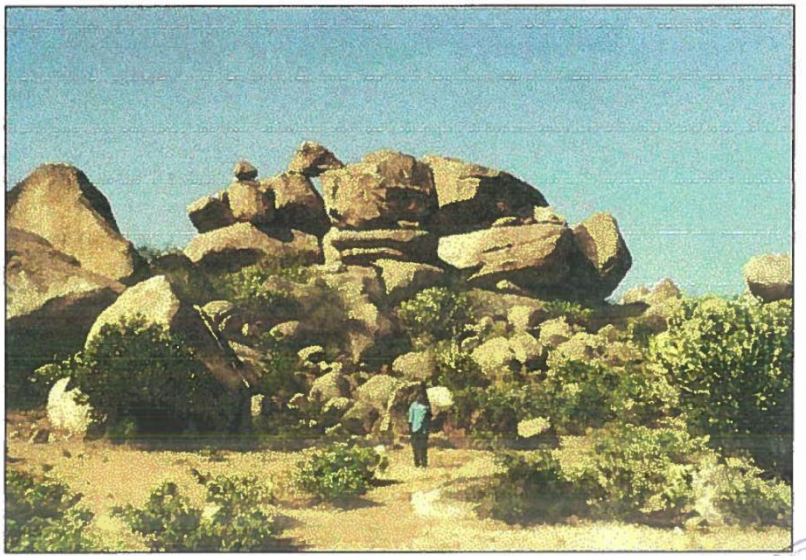


Photo: 1. Exposure of Chocolate brown Commercial Granite, Hasagal

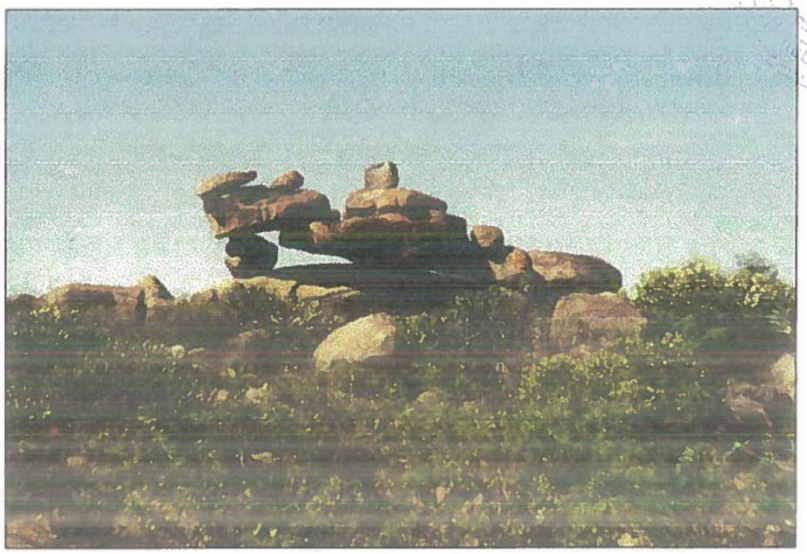
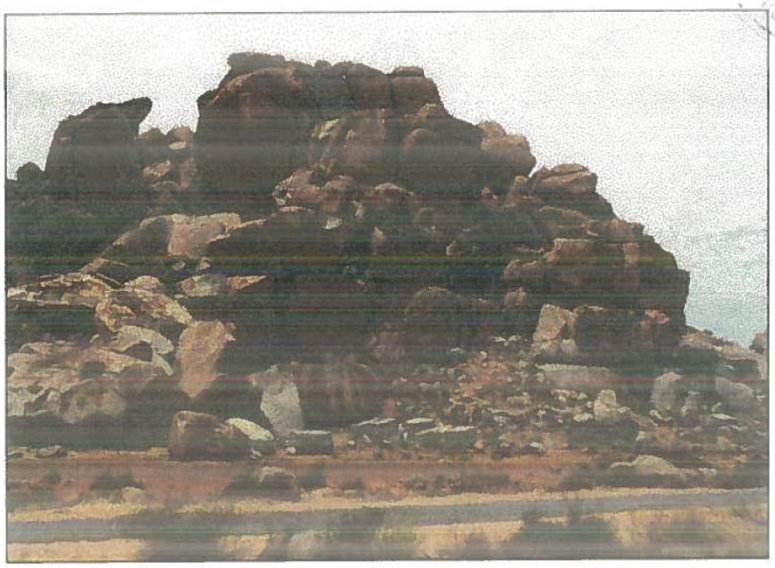


Photo: 2. View of Pink Phorphyry commercial Granite near Hire Sulikere.

Photo: 3. View of Chilly red Ornamental Granite Hill near Arasinakere



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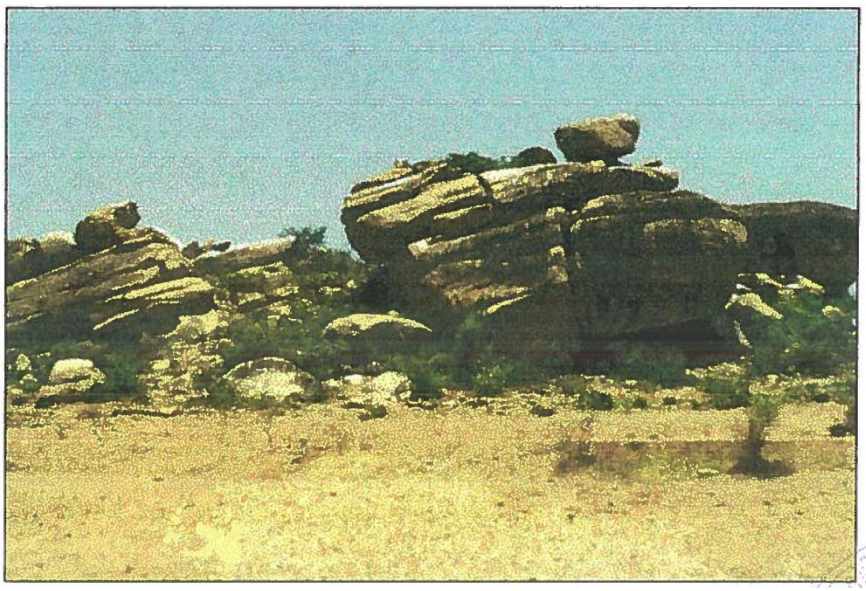

Photo: 5. Exposure of Pink porphyry commercial Granite exhibiting sub horizontal Joints and vertical joints near Hire Sulikere.

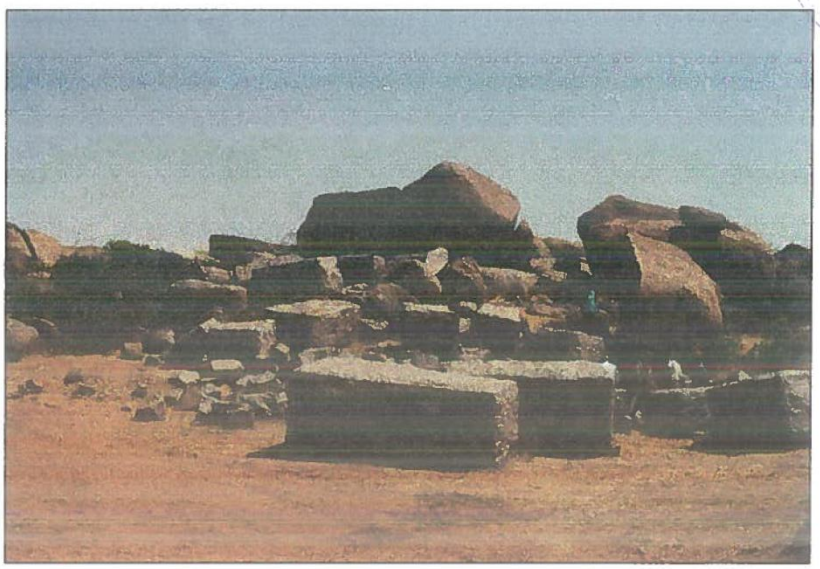

Photo: 6. View of gangue saw sized blocks of pink porphyry type ornamental Granite in a quarry located near Chik Sulikere Tanda.

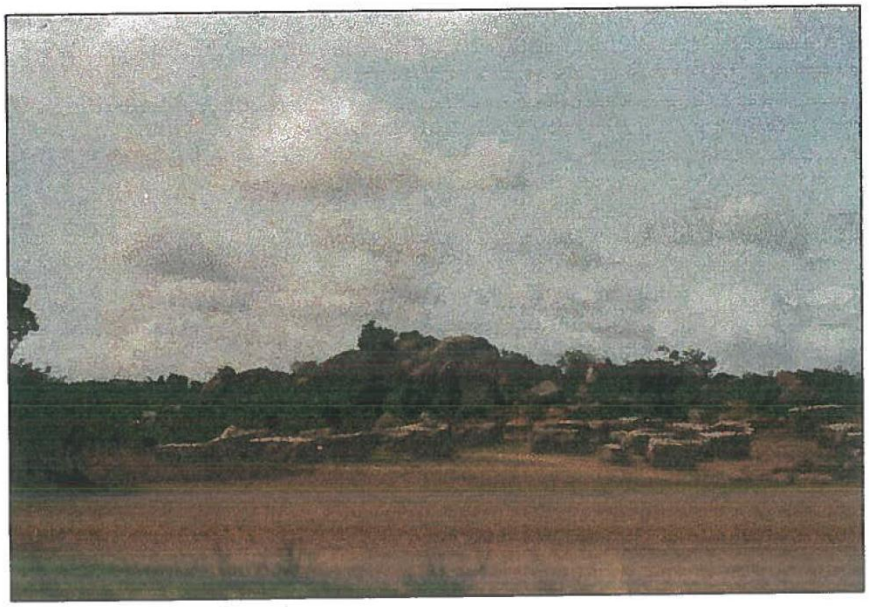

Photo: 7. Distribution of random size Chilly red commercial Granite in a quarry near Arasinakere.

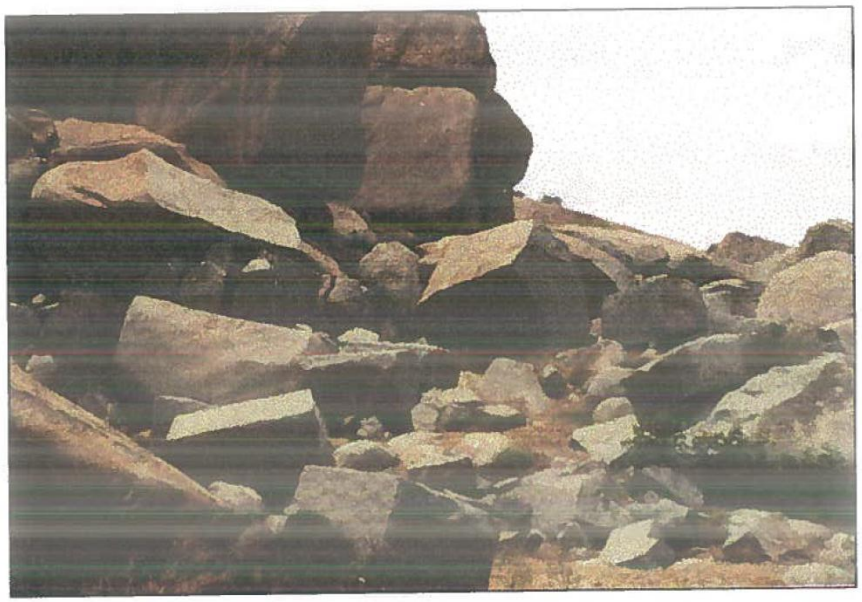

Photo: 8. A quarry face of Gangavathi grey commercial Granite near Venkatagiri.

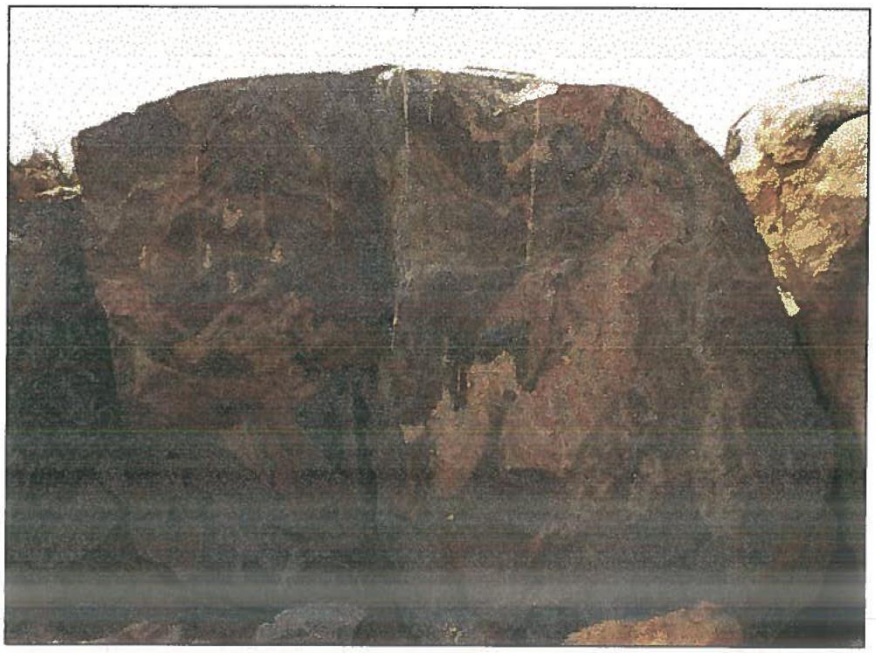

Photo: 9. An abandoned defective quarry of multicolour Granite near Hosagudda.

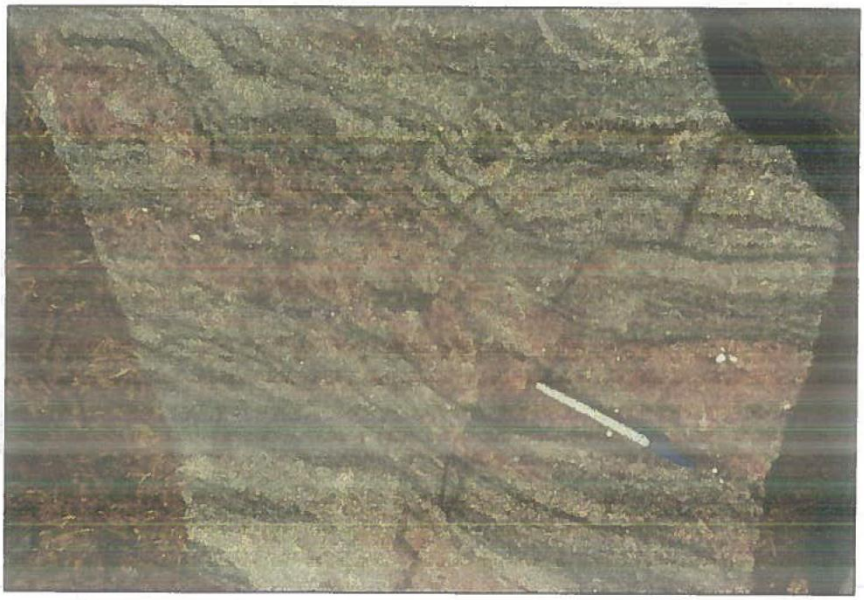

Photo: 10. A defective block of multicolour Granite near Hosagudda.

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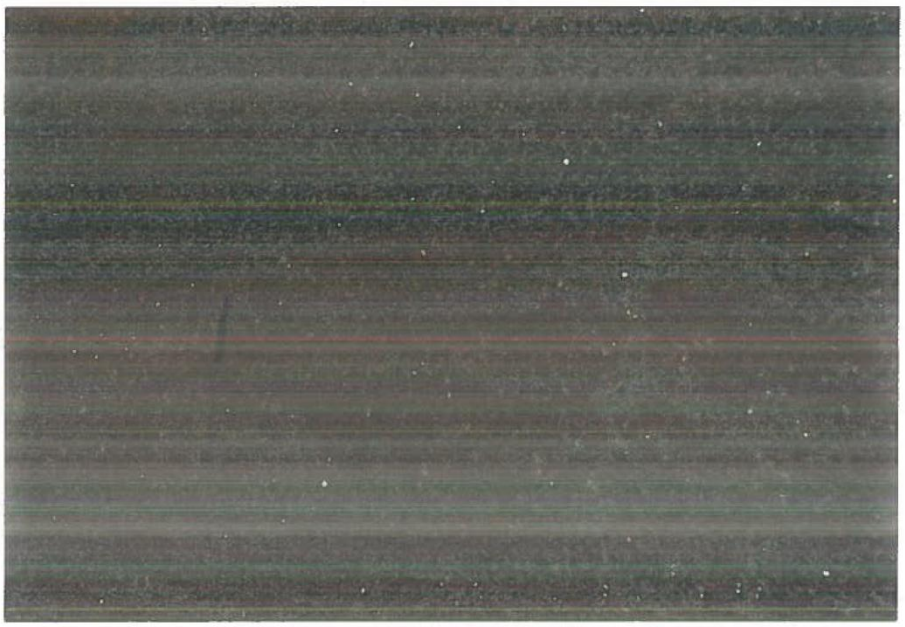

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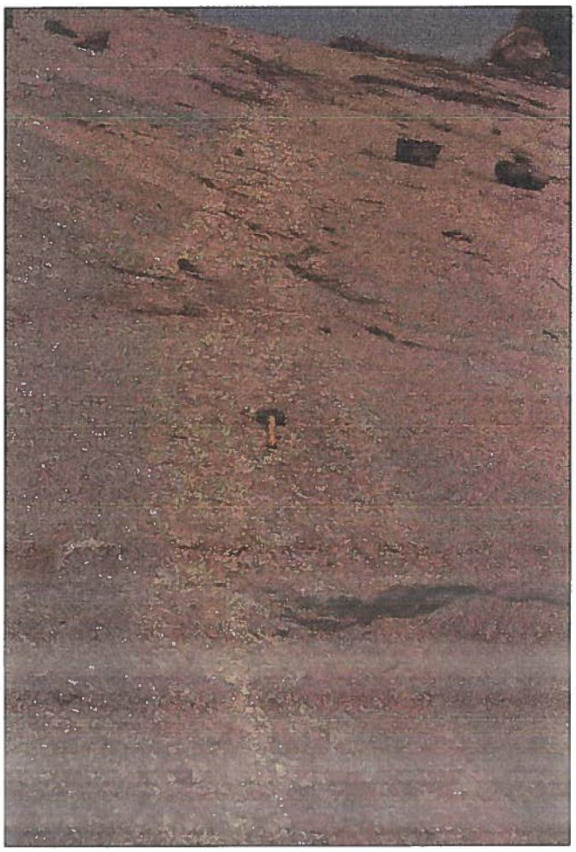

Photo: 13. View of wide pink Pegmatite with the pink porphyritic Granite in the quarry near Hosur

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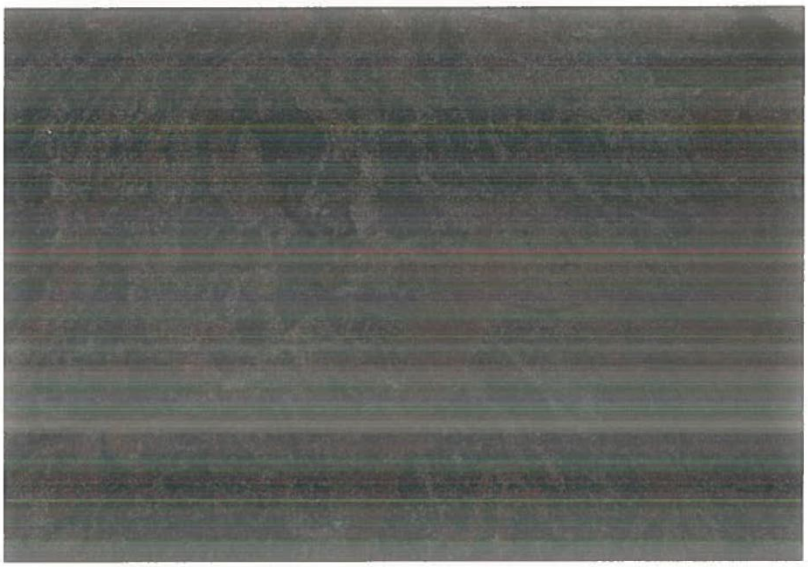

Photo: 14. Exposure of multicolour Granite showing crisscross pink veination near Hosur.

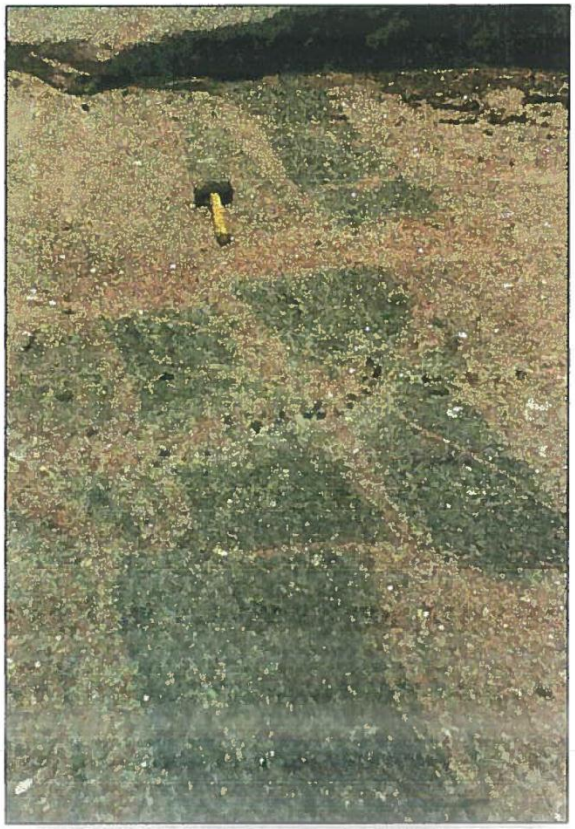

Photo: 15. Exposure of pink Granite dislocate and enclose the older grey granite converting in to multicolour granite.

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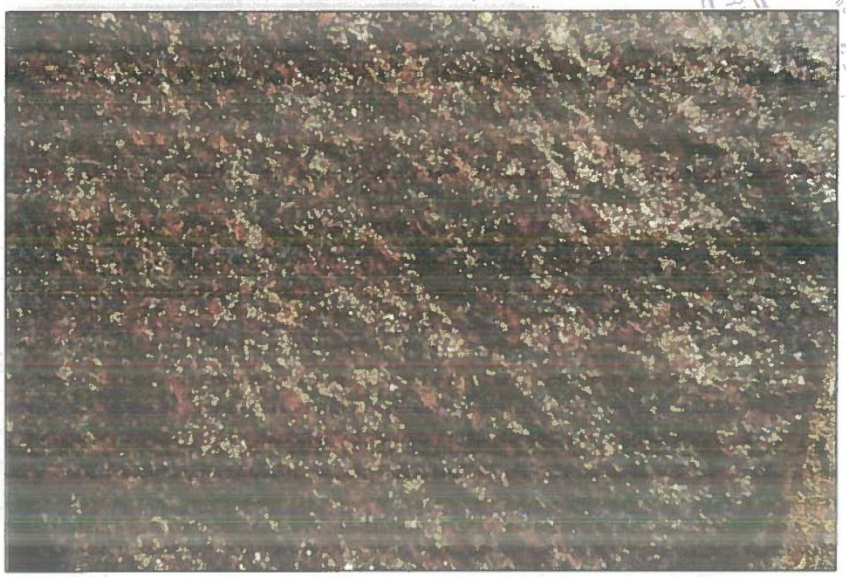

Photo: 16. Surface view of dressed gangue saw sized block of Sulikere red porphyry block (Radiant red) near Sulikere Tanda

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Photo: 18. An abandoned defective gangue saw size block of multicolour Commercial Granite near Hosagudda.
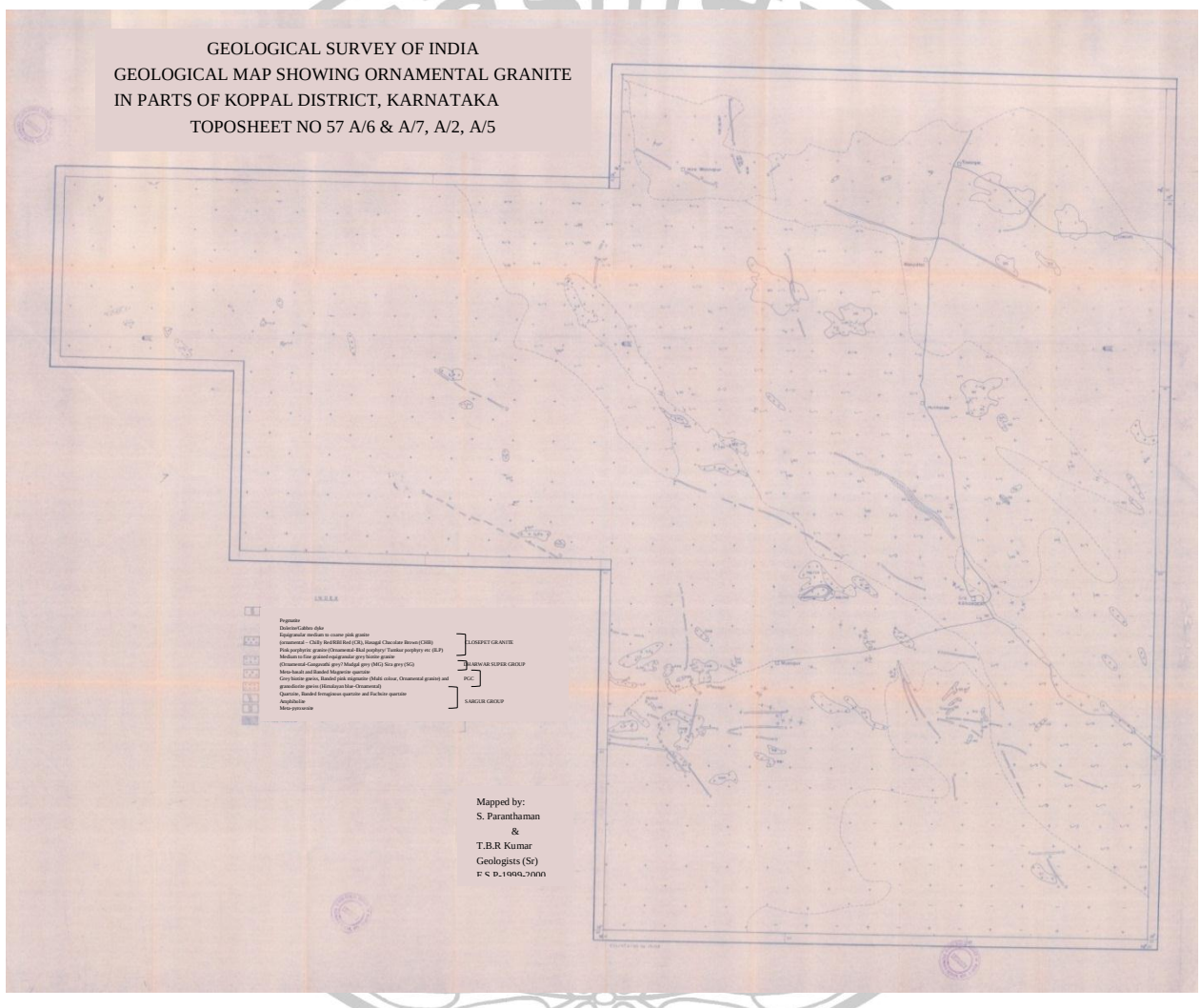
PLATE-I GEOLOGICAL MAP SHOWING ORNAMENTAL GRANITE IN PARTS OF KOPPAL DISTRICT, KARNATAKA